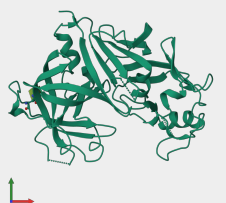


DEEPScreen

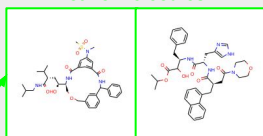
Wanming He, Stephen Marriott,
William Welsh

Introduction

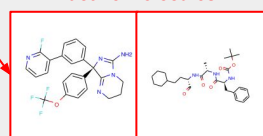
Protein Target (e.g.
Human Renin)



Active Molecules



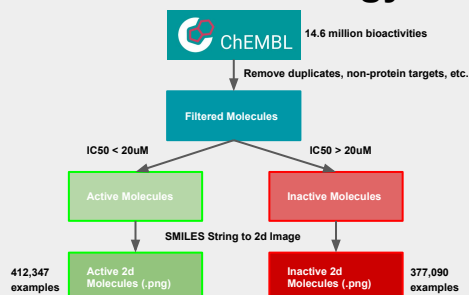
Inactive Molecules



Millions of potential drug candidates for specific therapeutic targets

In-silico screening can drastically lower costs by eliminating inactive molecules without physical testing

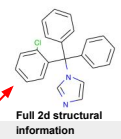
Data and Methodology



Example SMILES-to-image: Clotrimazole

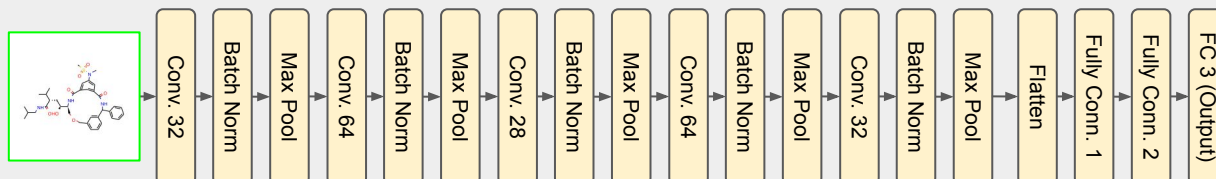
"Clc1ccccc1C(c1ccccc1)(c1ccccc1)n1ccnc1"

CNN

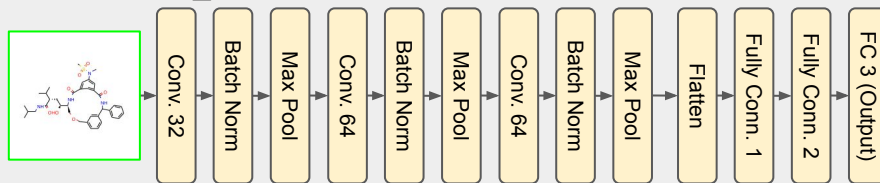


Model Architectures

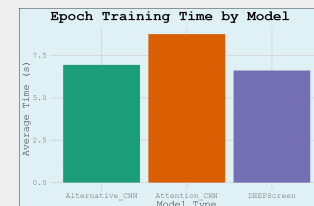
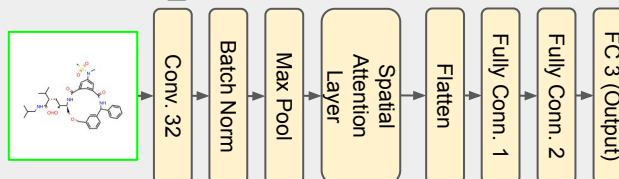
DEEPScreen



Alternative_CNN

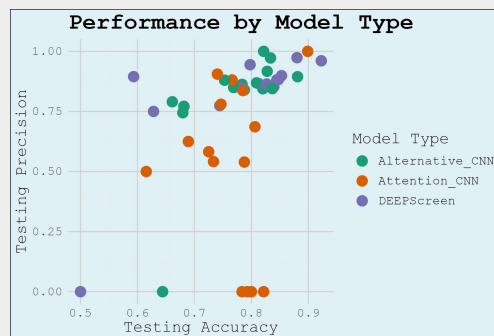


Attention_CNN



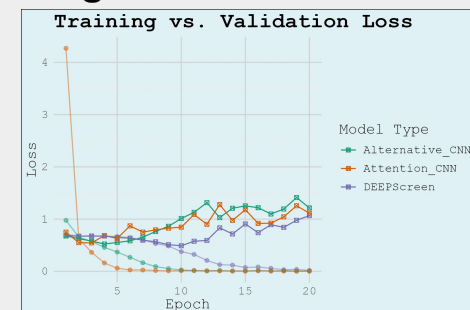
Despite being the least complex model by number of layers, the model incorporating spatial attention generally took longer to train.

Results



The three model architectures were tested on 15 example protein targets. By accuracy, the alternative CNN architecture we proposed performed slightly better than DEEPScreen on the preprocessed proteins we examined. The spatial attention model usually performed more poorly and had a wider variation in results.

Challenges and Future Work



The primary challenge faced by the alternative models we proposed was overfitting the training data. From the above chart, we can see training loss, represented with circles, decreases far more rapidly than for DEEPScreen, and diverges from validation loss much earlier. While we spent time attempting to modify our architecture and hyperparameters to avoid this, more work could be done to slow the rate at which these models overlearn from the training data. We think that there is promise in our two alternative approaches, provided more time spent tuning, given how well the two models learned the distribution of the data.